

Sameer Panghal

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

Expected Graduation Date: Spring 2026

M.S. Mechanical Engineering (*Concentration in Automation and Controls*)

Georgia Institute of Technology, Atlanta, GA

Aug 2022 – May 2025

B.S. Mechanical Engineering (*Graduated with Highest Honors*)

WORK EXPERIENCE

Undergraduate/Graduate Teaching Assistant – ME 2110 Atlanta, GA

Aug 2023 – Present

- Mentored roughly 70 students through the rigorous design-build-test cycle of electromechanical systems, teaching best practices in CAD, prototyping, and controls; this hands-on technical guidance helped drive five teams to top-8 finishes and one team to 1st place in the final competition.
- Served as a primary liaison between instructors and students across four lab sections, bridging conceptual gaps and clarifying complex deliverables while providing over 15 hours per week of dedicated support to ensure student alignment and success.
- Cultivated essential engineering "soft skills" by evaluating and critiquing over 100 design reports and project presentations, teaching students how to effectively structure arguments, deliver technical data, and master professional writing and public speaking.

Graduate Researcher – Advanced Crane Laboratory Atlanta, GA

May 2025 – Aug 2025

- Developed a nonlinear dynamic model of robotic crane's payload motion using Lagrangian mechanics and MATLAB symbolic toolbox, achieving model accuracy of 74% in simulation validation.
- Designed and implemented advanced input shaping and anti-sway control algorithms to minimize payload oscillations and enable precise level luffing
- Upgraded crane safety and control reliability by replacing 3D-printed parts with machined aluminum components and installing electrical stops, tip-over warning lights, power indicators, and warning labels to ensure operator awareness and hazard mitigation.

Robotics Intern – Schaeffler Group (prev. Vitesco Technologies), Seguin, TX

May 2024 – Aug 2024

- Designed, fabricated, and deployed End-Of-Arm-Tool (EOAT) to be mounted on Collaborative Robots (Co-bots) used for product pick and place applications capable of precisely handling loads of 2.5+ kg through 6 degrees of freedom utilizing SOLIDWORKS and Fused Deposition Molding (FDM)/Stereolithography (SLA) 3D printers.
- Simulated force loading on gripper and robot using Finite Element Analysis (FEA). Further developed design to decrease cantilever beam deflection by 3° by changing material, adding structurally significant bolting positions, and adjusting EOAT component geometric mating.
- Developed extensive drawings of EOAT components to be professionally machined as MIC6-Aluminum parts for dimensional stability and inertia minimization using SOLIDWORKS Drawing Tool, Geometric Design & Tolerancing (GD&T) Principles, and Design for Manufacturing/ Assembly Principles.
- Programmed Techman TM12S vision robot and Universal Robotics UR10e coupled with Keyence Ultrasonic Sensor to perform precise loading of customer products into vertical finished product racks for 3 separate customer products ranging in size, shape, geometry, and mass autonomously whilst optimizing placement using vision sensing to detect rack vacancy cutting head count expenses saving \$720,000 per year using TMFlow, TMVision, and Polyscript.
- Programmed UR10e Collaborative Robot to support Ford Automotive Electronic Control Unit production line using Polyscript.

Repair Process Engineer Intern – Delta TechOps, Atlanta, GA

Jan 2024 – May 2024

- Supported Stator and Vane Part Inspection and Repair Line performing cyclic loading analysis through 60+ aircraft engine components

- Developed 5 Engineering Repair Authorizations for unseen conditions found on engine parts saving \$1.2 million in replacement costs.
- Researched and implemented a new acidic treatment for a stained titanium engine stator component decreasing part damage cases by 13%.
- Developed new front-end software for process engineers to track all part cases and gain insight into team metrics using Teradata, DB Visualizer, Excel, Access, and Power Bi.

Undergraduate Researcher – LIDAR Lab, Atlanta, GA

May 2023 – Dec 2023

- Accomplished 300% gain in applied force by optimizing elliptical foot placement logic for bipedal robot CASSIE to avoid self-collision and recovery from applied forces utilizing MATLAB, Simulink, and inverse kinematic analysis.
 - Improved tactile sensing gripper design and reduced gripper size by 20% for humanoid robot DIGIT in SOLIDWORKS for better synchronization with GelSight sensors.
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PROJECT EXPERIENCE

Teleoperated Robotic Arm

Aug 2025 – Dec 2025

- Engineered a 4-DOF robotic arm utilizing custom-built actuators with 10:1 planetary gearboxes; performed static load calculations using MATLAB on worst-case configurations with a 1.3 safety factor to define torque requirements and validated the design by evaluating multiple gear material candidates to achieve a holding torque of ~18.8 N-m.
- Implemented closed-loop Field-Oriented Control (FOC) with PID tuning on ODrive S1 units and an STM32 NUCLEO-L476RG via CANBUS to execute teleoperated motion with 16-bit encoder precision
- Engineered a 600W power distribution and communication architecture using KiCad and daisy-chained CAN topology to drive three 20A continuous-current actuators and a 4th-axis gimbal.
- Developed a teleoperation pipeline using Python/Jupyter Notebooks to map computer vision model outputs directly to actuator setpoints, executing synchronized position control via C/C++ firmware managed on GitHub.

Tower Crane Vibration Reduction

Aug 2025 – Dec 2025

- Engineered an adaptive RLS control algorithm for a bridge crane by deriving a linearized discrete-time model, which reduced payload oscillations by ~85% during hardware validation
- Tuned the estimator's forgetting factor to $\lambda = 0.999$ to filter high sensor noise, enabling the system to accurately track and adapt to 20% changes in pendulum length (1m to 0.8m) during operation.
- Implemented Zero Vibration Derivative (ZVD) shaping using a zero-damping assumption to counteract unreliable hardware feedback (ζ approx. 0.4), ensuring consistent vibration suppression despite estimation errors.

Variable Input Hand Controlled Ball Levitation

Jan 2025 – May 2025

- Built a closed-loop mechatronic system integrating real-time sensing, actuator control, and signal processing to achieve stable levitation of a ball at any commanded height.
- Engineered physics-based plant model from first principles and experimental data, enabling accurate control law design and system performance prediction with model accuracy ~95%.
- Developed PD-based height control that reduced oscillations and maintained position within 2 mm of target, outperforming P and PID implementations in stability and responsiveness.
- Applied noise mitigation using 10-point moving average, Butterworth, and Kalman filters to enhance measurement reliability.
- Overcame nonlinear system limitations through hardware-software co-design, including optimal placement of ultrasonic sensors, servo-actuated throttle control, and integration of brushless DC fan airflow with safety-critical hardware.

STRIDE School Bus Navigation Device (CREATE-X)

Jan 2025 – May 2025

- Designed and validated a ruggedized HUD enclosure using PLA and TPU additive manufacturing, utilizing MATLAB/Simulink for vibration analysis to achieve a natural frequency >300 Hz and ensure structural integrity under simulated 10G crash loads.

- Optimized optical system geometry using geometric ray-tracing calculations to define a 5.2-inch screen distance and semi-reflective combiner path, achieving a 2000 mm Virtual Image Distance (VID) and 15,000 nits brightness for daylight legibility.
- Validated electromechanical system performance using Raspberry Pi integration and Python-based data analysis, comparing on-road field test data against Google Maps API baselines to confirm ± 5 -meter GPS accuracy and ETA predictions within ± 17 seconds.
- Secured a summer pilot program agreement with Atlanta Public Schools (APS) using a functional MVP demonstration and cost-benefit analysis presented to the Deputy Transportation Head, projecting improved routing reliability for 40,000+ students and a $\sim 38\%$ reduction in recurring fleet telemetry costs.

Motor Testing/Storage Jig

Aug 2022 – Present

- Designed motor storage jig for HT07 EV Racecar's electric motors, supporting 4 AMK package servo motors. Supported mass of 25+ pounds and maximum torque output of 84 Nm in CATIA.
- Simulated torque and force loading on motor jig using Finite Element Analysis (FEA) in CATIA. Developed structural improvements leading to a 15% increase in force tolerance capabilities.
- Fabricated motor jig using CAD along with CNC and welding techniques. Completed structure to hold 4 motors at a 90° angle while running 50+ simultaneous tests.

IG-11 Competition Robot

May 2023 – July 2023

- Developed an autonomous robot for high-intensity competitions, excelling in retrieval, deposition, and toppling tasks with size, time, power, and financial constraints. Detailed progress through technical reports and presentations. Attained 2nd highest score overall.
- Conceptualized robot design using Quality Function Deployment (QFD) tools including House of Quality, Specification Sheet, Morphological Chart, and Function Tree.
- Fabricated physical robot through woodworking, laser cutting using Inkscape, and 3D Printing using Cura Ultimaker. Developed custom robot logic using Arduino to control solenoids, motors, pneumatics, and mechanical switches in C++ ensuring efficient actuator control.

SKILLS

- Design Skills: Computer Aided Design (CAD), SOLIDWORKS, CATIA, Onshape, Cura Ultimaker, Bambu Studio, Inkscape, Microsoft Suite, Adobe Studio, Robotics, Geometric Dimensioning & Tolerancing (GD&T), Motor Sizing
- Programming Skills: MATLAB, Simulink, Java, Python, SQL, C, C++, DMA, I2C, Serial Communication, Git, ROS, Raspberry Pi
- Fabrication Skills: CNC Machining, Welding, 3D Printing, Laser Cutting, Power Tools, Bandsaw, Dremmel